



SOLIDITY 101

An Introduction to Solidity : Session – 01/03

Web3Assam Winter Cohort 2025



Contents

Session Length: 1.5 Hours

- ☐ **Smart Contract**
- ☐ **Ethereum Basics**
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- ☐ **Gas & Transactions**
- ☐ **Smart Contract Lifestyle**
- ☐ **Bytecode & ABI**
- ☐ **Basic Intro to Solidity**

Smart Contract



- ❑ A program stored and executed on the blockchain.
- ❑ Runs exactly as written - no downtime, no manual control, no third party.
- ❑ Cannot be changed once deployed (immutable).
- ❑ Holds data and logic just like a backend server - but decentralized.

Smart Contract = Code + State + Rules

- ✓ Code → functions
- ✓ State → variables stored on-chain
- ✓ Rules → enforced automatically

Why Smart Contracts Exist

- Remove intermediaries
- Enable automation
- Provide transparency
- Guarantees correctness (if code is correct)

How Smart Contracts Work (Internally)

- Written in Solidity
- Compiled into bytecode
- Deployed on Ethereum blockchain
- Executed by the EVM (Ethereum Virtual Machine)
- Readable by any user or application

ETHEREUM



Ethereum History

- ✓ Ethereum was conceptualized by Vitalik Buterin, a developer and writer at Bitcoin Magazine.
- ✓ Vitalik believed Bitcoin's scripting language was too limited.
- ✓ He proposed a blockchain with a fully programmable smart contract platform.
- ✓ Wrote the Ethereum Whitepaper in late 2013, introducing the idea of a decentralized world computer.
- ✓ Core motivation: *Blockchains should do more than send money. They should run decentralized applications.*

- ✓ Team Formation (Early 2014) Ethereum attracted early co-founders:

- ☐ Vitalik Buterin
- ☐ Gavin Wood (wrote the Yellow Paper, defined the EVM)
- ☐ Joseph Lubin (later founded ConsenSys)
- ☐ Mihai Alisie
- ☐ Anthony Di Iorio
- ☐ Charles Hoskinson (later founded Cardano)
- ☐ Amir Chetrit

- ✓ Gavin Wood's Yellow Paper formally defined the Ethereum Virtual Machine (EVM) and became the technical foundation.

Ethereum as the World Computer 🚀



Ethereum is often called “The World Computer” because it acts like a global, decentralized machine where anyone can deploy programs (smart contracts) that run exactly as written - without downtime, censorship, or third-party control.

What Does “World Computer” Mean?

Ethereum as a single virtual computer running on thousands of machines worldwide (nodes). Every node contributes to:

- ❑ Storing the blockchain data
- ❑ Running smart contracts
- ❑ Verifying transactions
- ❑ Enforcing rules of the Ethereum protocol

Together, they form one giant distributed computer that never stops.

➡ Every node runs the *same code* → *same result* → *global consensus*.

Why Ethereum Behaves Like a Computer?

Ethereum exposes fundamental computer components:

Computer Part	Ethereum Equivalent
CPU	Ethereum Virtual Machine executes code
Hard Disk	Blockchain Storage stores contract state
RAM	EVM Memory + Stack used during execution
Programs	Smart Contracts (Bytecode)
Inputs	Transactions / Function Calls

Ethereum History

What Is the EVM? (Ethereum Virtual Machine)

- ✓ The EVM is the engine that executes all smart contract code.
- ✓ Deterministic → same input = same output
- ✓ Isolated → contracts cannot access OS, internet
- ✓ Sandboxed → cannot harm the host
- ✓ Gas-metered → prevents infinite loops
- ✓ All nodes run the EVM → ensures global agreement on results.

This is why Ethereum is considered a trustless, decentralized computer.

Smart contracts are:

- ✓ Immutable programs
- ✓ Deployed once to Ethereum
- ✓ Executed by the EVM
- ✓ Accessible to anyone
- ✓ Once deployed : No server is needed
- ✓ No one can modify or delete them
- ✓ They run exactly as written

This is similar to uploading a program to a shared computer that the whole world must run exactly the same way.

Ethereum Gas

Gas: The “Fuel” of the World Computer

Gas is how Ethereum measures computational cost. In normal computers, your CPU does work for free. In Ethereum, computation has cost:

- ☐ More storage → more gas
- ☐ More complex computation → more gas
- ☐ Infinite loops → impossible (would run out of gas)

Gas serves two purposes:

- ☐ Prevents abuse (like infinite loops, spam)
- ☐ Pays miners/validators for computation

Gas is paid in ETH.

Bytecode: What the EVM Actually Executes

Your Solidity code is not directly executed. Flow:

- ☐ Write Solidity
- ☐ Compiler → converts into EVM bytecode
- ☐ Bytecode is stored on-chain
- ☐ Nodes execute this bytecode inside the EVM

Bytecode is the “machine code” of the world computer.

ABI: Application Binary Interface

ABI is how external users, apps, or contracts talk to your smart contract. ABI ensures:

- ☐ How functions are encoded
- ☐ How arguments are packed
- ☐ How return values are decoded

Without ABI → nobody knows how to call your contract.

Ethereum Networks

- ❑ **Mainnet:** Official, real Ethereum blockchain. It uses real ETH with real monetary value. Transactions cost real gas and are irreversible. It is used only for final, production-ready smart contracts. It has highest security, fully decentralized, globally validated.
- ❑ **Testnet:** Used for testing without risking real ETH.

Sepolia :

- Primary testnet for most development.
- Stable, widely supported, fast confirmations.
- Free test ETH from faucets.

Holesky :

- Designed for large-scale testing and staking scenarios.
- Massive test ETH supply for validator/infra testing.
- More suitable for system-level experiments.

- ❑ **Devnet (Local Development Networks):** Local blockchains running on a developer's computer.

Ethereum Account

An Ethereum account is an object stored in Ethereum's state that contains:

- ☐ Nonce – number of transactions sent
- ☐ Balance – amount of ETH owned
- ☐ StorageRoot – storage data (only for smart contracts)
- ☐ CodeHash – contract code (empty for regular users)

Every account has an Ethereum address (20 bytes).

Feature	EOA (User)	Contract Account
Controlled By	Private key	Smart contract code
Can initiate tx?	✓ Yes	✗ No
Has bytecode?	✗ No	✓ Yes
Holds ETH?	Yes	Yes
Storage?	No	Yes

Types of Wallets

- ☐ Custodial Wallet & Non-Custodial Wallet
- ☐ Software Wallets & Hardware Wallets
- ☐ Paper Wallet & Smart Contract Wallets
- ☐ Hot Wallets & Cold Wallets

Specialized Types of Wallets

- ☐ Multi-Signature Wallets (Multisig)
- ☐ MPC Wallets (Multi-Party Computation)
- ☐ Social Recovery Wallets
- ☐ Embedded Wallets

Assignment

- ❖ Create a Wallet on Metamask
- ❖ Use a Faucet to get some test ETH
- ❖ Send a test transaction on Sepolia
- ❖ Complete the Basic Syntax Series of Solidity in YouTube

Thank You